Cardiovascular System (CARD) · NPLEX-P1-CARD-06

CV-6 Cardiac Output and Its Regulation

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-06-01-C01-BD1

Brain dump 1 · foundational retrieval: Write out the output and pressure equations

A competent response contains

· CO = SV x HR; resting CO is about 5 L/min at an SV of about 70 mL and a rate of about 72 bpm.

· SV = EDV - ESV; a resting left ventricle holds an EDV of about 120 mL and an ESV of about 50 mL.

· EF = SV / EDV, normally about 55 to 70%.

· Cardiac index is CO divided by body surface area, about 2.5 to 4 L/min per square metre.

· CO = (MAP - RAP) / TPR, and with RAP near 0 mmHg this becomes MAP = CO x TPR; normal TPR is about 18 mmHg per L/min and normal MAP about 70 to 100 mmHg.

· From a cuff, MAP is about diastolic pressure plus one third of the pulse pressure, so 120/80 mmHg gives about 93 mmHg.

· The one third weighting reflects diastole filling about two thirds of the cycle at resting rates; at fast rates diastole shortens more than systole, so the rule underestimates MAP.

· Fick: CO = VO2 / (CaO2 - CvO2), with VO2 about 250 mL/min at rest from expired gas analysis.

· O2 content = (1.34 x hemoglobin in g/dL x fractional saturation) + (0.003 x PO2 in mmHg), in mL O2 per dL; the first term is hemoglobin-bound and the second is dissolved, and dissolved O2 adds only about 0.3 mL/dL.

· Multiply content in mL/dL by 10 to express it per litre.

· Arterial content can be sampled from any systemic artery, but mixed venous content must come from the pulmonary artery, because venous blood from different organs differs widely in saturation.

Distinctions that matter

· Stroke volume is a per-beat volume and cardiac output is a per-minute flow, so a change in rate alters output without altering stroke volume.

· Ejection fraction is a ratio and partly corrects for chamber size, while stroke volume is an absolute volume.

Common misconceptions

· Taking MAP as the average of systolic and diastolic pressure. Diastole occupies about two thirds of the resting cycle, so the weighting is toward diastolic pressure.

· Leaving the dissolved term out of O2 content as if it were zero. It is small at about 0.3 mL/dL but it is part of the formula.

Where it is taught: Section C01, opening paragraphs and the Fick principle paragraphs of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C01-BD2

Brain dump 2 · integration and application: Work one patient two ways and read the venous saturation

A competent response contains

· SV is 150 minus 90, which is 60 mL, below the normal 70 mL.

· CO is 60 mL times 60 bpm, which is 3.6 L/min, below the normal 5 L/min.

· EF is 60 divided by 150, which is 40%, below the normal 55 to 70%.

· CaO2 = (1.34 x 12 x 0.97) + (0.003 x 90) = 15.6 + 0.27, about 15.9 mL/dL.

· CvO2 = (1.34 x 12 x 0.55) + (0.003 x 30) = 8.8 + 0.09, about 8.9 mL/dL.

· The arteriovenous difference is about 7.0 mL/dL, which is 70 mL O2 per litre of blood.

· CO = 240 / 70, which is about 3.4 L/min, close to the volumetric value of 3.6 L/min.

· A mixed venous saturation of 55% is low, so tissues are extracting more O2 from each litre and delivery is inadequate for demand.

· The arteriovenous difference widened because VO2 stayed the same while CO fell, and the Fick equation allows only those two ways to balance.

· A forearm vein samples one bed only and is usually less desaturated than true mixed venous blood, so the calculated difference would be too narrow and the calculated output too high.

Distinctions that matter

· Reduced output with a normal VO2 widens the arteriovenous difference, while a raised VO2 at a normal output widens it as well, so the two must be separated by knowing VO2.

· Low mixed venous saturation reports the adequacy of delivery relative to demand, not the arterial oxygenation, which is normal at 97% here.

Common misconceptions

· Dividing VO2 by a content difference still expressed per decilitre. That gives an answer ten times too large.

· Treating a low ejection fraction and a low cardiac output as the same finding. A dilated ventricle can hold a near-normal stroke volume at a low ejection fraction by filling to a very large EDV.

Where it is taught: Section C01, Fick paragraphs and the worked example on calculating output by two routes of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C02-BD1

Brain dump 1 · foundational retrieval: Draw the Frank-Starling curve and define the three determinants

A competent response contains

· The curve of SV against EDV rises steeply and then flattens, and the normal heart works on the ascending limb.

· Further stretch is resisted by stiff myocardium and by the pericardium.

· Raised contractility shifts the whole curve up and to the left, giving a larger SV at the same EDV; reduced contractility shifts it down and to the right.

· Preload is strictly end-diastolic wall stress, indexed in practice by EDV, and is the stretch applied to the muscle before it contracts.

· Afterload is systolic wall stress, approximated by aortic pressure, and is the load the ventricle must overcome to open the aortic valve and eject.

· Contractility is the strength of contraction at a fixed preload and afterload, indexed by the slope of the end-systolic pressure-volume relationship, and is set by myofilament Ca2+ delivery and responsiveness.

· End-diastolic pressure, estimated for the left ventricle by pulmonary capillary wedge pressure, tracks volume only when compliance is normal, so a stiff ventricle shows a high filling pressure at a normal EDV.

· Optimal sarcomere length is about 2.2 micrometres.

· Stretch increases force by improving filament overlap and by raising the Ca2+ sensitivity of troponin C, and the Ca2+ sensitivity effect matters more.

· The Frank-Starling mechanism also keeps right and left ventricular outputs matched beat to beat.

Distinctions that matter

· Moving along one curve is a preload change; moving to a different curve is a contractility change.

· Preload acts through EDV, while afterload and contractility act through ESV.

Common misconceptions

· Treating filling pressure and filling volume as interchangeable measures of preload. They separate as soon as compliance falls.

· Explaining the Frank-Starling mechanism by filament overlap alone. Length-dependent Ca2+ sensitivity of troponin C is the larger part.

Where it is taught: Section C02, paragraphs on preload and the Frank-Starling mechanism and on contractility of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C02-BD2

Brain dump 2 · integration and application: Follow three single changes through to stroke volume

A competent response contains

· Raised aortic pressure acts on afterload first: more of each contraction is spent building pressure, so ESV rises and SV and EF fall, the Frank-Starling curve shifts down and right, and the loop becomes taller and narrower with the end-systolic point moving right along the same end-systolic pressure-volume relationship.

· Over the next beats the residual blood adds to normal filling, EDV rises, and the Frank-Starling mechanism restores much of the lost SV, at the cost of a higher filling pressure.

· A litre of saline acts on preload first: EDV rises, ESV is about unchanged, SV rises and EF rises slightly, the heart moves up the same Frank-Starling curve, and the loop widens to the right.

· A coronary occlusion that leaves part of the ventricle contracting weakly acts on contractility first: ESV rises, SV and EF fall, the Frank-Starling curve shifts down and right with a flatter end-systolic pressure-volume relationship, and EDV then rises as residual volume accumulates.

· EF is not a clean index of contractility because it also falls when afterload rises, and it can look near normal in a dilated ventricle with a very large EDV.

· At 200 bpm the extra beats come mostly out of diastole, so filling time shortens, EDV falls, and SV and then CO fall.

· Sympathetic lusitropy and inotropy raise the rate at which output starts to fall, while a stiff ventricle that depends on atrial contraction loses output at lower rates.

Distinctions that matter

· Afterload and reduced contractility both raise ESV, and they are separated by whether the end-systolic pressure-volume relationship itself has changed slope.

· The immediate effect of a change and the net effect after several beats of Frank-Starling adjustment are different answers to different questions.

Common misconceptions

· Expecting a faster rate always to raise cardiac output. Beyond a point the loss of filling time outweighs the extra beats.

· Reading a fall in ejection fraction as proof that contractility has fallen. A rise in afterload lowers ejection fraction with contractility unchanged.

Where it is taught: Section C02, afterload and contractility paragraphs, the prediction table, and the stop and predict box of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C03-BD1

Brain dump 1 · foundational retrieval: Draw the cardiac function and venous return curves

A competent response contains

· The cardiac function curve is the Frank-Starling curve with RAP as the measure of filling: CO rises steeply with RAP and then plateaus.

· The venous return curve falls as RAP rises, and the crossing point is the operating point, where CO equals venous return.

· The two flows must be equal in the steady state because the heart can pump only what returns to it.

· The venous return curve meets the horizontal axis where RAP equals mean systemic filling pressure, about 7 mmHg, which is the pressure that would exist throughout the systemic circulation if the heart stopped and pressures equalised.

· Venous return = (MSFP - RAP) / RVR, where RVR is the resistance to venous return, most of which lies in the veins.

· The slope of the curve is 1 divided by RVR.

· The curve plateaus once RAP falls below atmospheric pressure, because the large veins entering the thorax collapse.

· Stressed volume is the blood above what fills the vessels without stretching them, and only stressed volume generates pressure.

· The veins hold about two thirds of the blood.

· Venoconstriction converts unstressed volume into stressed volume, so MSFP rises and the curve shifts right, parallel to itself; transfusion does the same, while hemorrhage and venodilation shift it left.

Distinctions that matter

· A change in MSFP shifts the venous return curve parallel to itself, while a change in resistance rotates it about the same x-intercept.

· The cardiac function curve answers what the heart can do with a given filling pressure, and the venous return curve answers what the vasculature will deliver at that pressure.

Common misconceptions

· Treating total blood volume as what sets MSFP. Only the stressed portion generates pressure, which is why venous tone changes MSFP without any change in volume.

· Reading the venous return curve as falling without limit at low RAP. Venous collapse at the thoracic inlet flattens it.

Where it is taught: Section C03, opening paragraphs and the venous return curve paragraphs of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C03-BD2

Brain dump 2 · integration and application: Name the problem from cardiac output and right atrial pressure

A competent response contains

· When CO and RAP change in the same direction the primary problem is vascular, in volume or venous tone; when they change in opposite directions the problem is in the heart.

· Patient A: low CO with low RAP is inadequate filling, a vascular cause, with the venous return curve shifted left by a fall in MSFP, as in hemorrhage or venodilation.

· Patient B: low CO with high RAP is pump failure, with the cardiac function curve shifted down and right by reduced contractility.

· Patient C: three days of vomiting lowers blood volume and MSFP, shifting the venous return curve left, so CO and RAP both fall together, a vascular cause.

· Patient D: a large anterior infarction lowers contractility and shifts the cardiac function curve down and right, so CO falls while RAP rises, a cardiac cause.

· A fall in TPR rotates the venous return curve up about the same x-intercept and removes afterload so the cardiac function curve shifts up, so CO rises while RAP stays about the same.

· Because both curves move together, resistance changes show up as a change in flow with little change in filling pressure, which is why an arteriovenous fistula or a widely dilated muscle bed raises output.

Distinctions that matter

· The direction of RAP relative to CO is the clue that separates a cardiac cause from a vascular one.

· Volume and venous tone act on MSFP and move CO and RAP the same way; contractility moves them in opposite ways; TPR moves CO alone.

Common misconceptions

· Reading any low cardiac output as heart failure. A low RAP alongside it points to the vascular compartment instead.

· Expecting arteriolar constriction to raise RAP. It moves both curves in the same sense, so RAP changes little and CO falls.

Where it is taught: Section C03, the operating point table and the paragraph on the direction of right atrial pressure of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C03-RT01

Retrieval exercise: Rebuild the hemorrhage sequence on the curves

A competent response contains

· Step 1: blood loss reduces stressed volume.

· Step 2: MSFP falls and the venous return curve shifts left.

· Step 3: the new intersection sits at a lower RAP and a lower CO.

· Step 4: the lower MAP reduces baroreceptor firing.

· Step 5: sympathetic venoconstriction raises MSFP toward normal.

· Step 6: sympathetic inotropy shifts the cardiac function curve up.

· Step 7: CO partly recovers while RAP stays low.

· Blood loss lowers MSFP because it removes stressed volume, the blood above what fills the vessels without stretching them.

· The sensors are the carotid sinus and aortic arch baroreceptors.

· Venoconstriction restores the venous return curve by raising MSFP; inotropy restores the cardiac function curve by shifting it up and to the left.

· RAP stays low because the venous return curve is still shifted left and the upward shift of the cardiac function curve pulls RAP down further.

Distinctions that matter

· The primary event here is a shift of the venous return curve, and the reflex response moves both curves.

· Partial recovery of output is not recovery of volume; the operating point still sits at a low filling pressure.

Common misconceptions

· Expecting RAP to return to normal as output recovers. Reflex inotropy raises output by pumping the atrium emptier, so RAP falls rather than rises.

· Thinking a loss of about 10% of blood volume should drop arterial pressure sharply. The reflex holds pressure nearly constant at that point, so the operating point moves while the cuff reading barely changes.

Where it is taught: Section C03, flow diagram on hemorrhage, with the reflex detail in C05 of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C04-BD1

Brain dump 1 · foundational retrieval: Write the law of Laplace and the determinants of oxygen demand

A competent response contains

· Wall stress = (P x r) / (2 x h), where P is intraventricular pressure, r is chamber radius and h is wall thickness.

· Wall stress is the force each unit of wall must develop.

· In systole wall stress is the afterload, and it is a major determinant of myocardial O2 demand.

· Higher pressure raises wall stress in direct proportion.

· A larger radius raises wall stress in direct proportion, so a dilated ventricle must develop more force per unit of wall to produce the same pressure.

· A thicker wall lowers wall stress, because the load is shared among more muscle.

· Myocardial O2 demand is set mainly by heart rate, which is the number of tension-generating cycles per minute, by contractility, through the ATP used in Ca2+ cycling and cross-bridge activity, and by systolic wall stress.

· Developing pressure costs far more O2 than ejecting volume, so pressure work is much more expensive than volume work.

· The rate-pressure or double product is heart rate times systolic pressure.

· The heart already extracts about 70 to 75% of the O2 delivered to it at rest, so higher demand must be met almost entirely by higher coronary flow, and most left ventricular flow occurs in diastole.

Distinctions that matter

· Pressure and radius raise wall stress while thickness lowers it, so the same equation explains both the problem of dilation and the benefit of hypertrophy.

· Wall stress is force per unit of wall cross-section, not the pressure inside the chamber.

Common misconceptions

· Treating afterload as simply the aortic pressure. Aortic pressure is the practical approximation; the load itself is systolic wall stress, which also depends on radius and thickness.

· Assuming higher demand can be met by extracting more O2. Extraction is already about 70 to 75% at rest, so flow has to rise.

Where it is taught: Section C04, opening paragraph and the paragraphs on determinants of myocardial O2 demand of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C04-BD2

Brain dump 2 · integration and application: Remodeling and the ischemic cost of a faster rate

A competent response contains

· The first patient has pressure overload, signalled by raised systolic wall stress, with sarcomeres added in parallel so the wall thickens and the cavity stays normal or small, which is concentric hypertrophy.

· Raising systolic pressure from 120 to 200 mmHg increases wall stress by about two thirds, and thickening the wall from 1.0 to about 1.7 cm returns stress toward normal.

· The costs are a stiff chamber that needs a high filling pressure, which is diastolic dysfunction with an S4, and capillary growth that lags myocyte growth.

· Because capillary supply lags, the subendocardium becomes vulnerable to ischemia even with normal coronary arteries.

· Angiotensin II and catecholamines help drive the remodeling alongside the mechanical load.

· The second patient has volume overload, signalled by raised diastolic wall stress, with sarcomeres added in series so the chamber enlarges, which is eccentric hypertrophy.

· While compensation holds, the wall thickens enough to keep the thickness to radius ratio about normal, so wall stress stays near normal and the chamber delivers a large stroke volume; when it fails, radius grows without matching thickness, the ratio falls, and wall stress and the O2 cost of each beat rise in muscle that is already weakened.

· The third patient's demand rises because heart rate is one of the three main determinants of myocardial O2 demand, and the rate-pressure product doubles.

· On the supply side, most left ventricular coronary flow occurs in diastole, and a faster rate shortens diastole, so the time available for supply falls while demand rises.

Distinctions that matter

· Both remodeling patterns aim to restore normal wall stress; they differ in whether the overload acts in systole or in diastole.

· Concentric hypertrophy usually fails on the filling side with an S4, while decompensated dilation fails on the ejection side with an S3.

Common misconceptions

· Reading hypertrophy as purely harmful. The thicker wall lowers wall stress and is a corrective response; the harm comes from stiffness and from capillary supply lagging.

· Explaining tachycardia-induced ischemia on the demand side only. Shortened diastole cuts coronary supply at the same time.

Where it is taught: Section C04, paragraphs on remodeling under chronic overload and on myocardial O2 demand of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C04-CP01

Comparison table: Concentric against eccentric remodeling

A competent response contains

· Typical cause: concentric from pressure overload such as hypertension or aortic stenosis; eccentric from volume overload such as aortic or mitral regurgitation, and from a failing dilated ventricle.

· Signal: concentric follows raised systolic wall stress; eccentric follows raised diastolic wall stress.

· Sarcomere addition: in parallel, giving wider cells, in concentric hypertrophy; in series, giving longer cells, in eccentric hypertrophy.

· Thickness to radius ratio: increased in concentric hypertrophy; about normal in compensated eccentric hypertrophy and decreased in decompensated dilation.

· Wall stress: returns toward normal in concentric hypertrophy; near normal while eccentric hypertrophy is compensated, then rises as the chamber dilates.

· Chamber stiffness and filling pressure: concentric gives a stiff chamber needing a high filling pressure; eccentric keeps compliance higher and accommodates a large volume until it decompensates.

· Typical consequence: concentric gives diastolic dysfunction with an S4; eccentric gives a large stroke volume when compensated and systolic dysfunction with an S3 when decompensated.

Distinctions that matter

· The thickness to radius ratio is the quickest discriminator, because it rises in concentric hypertrophy and holds or falls in eccentric hypertrophy.

· Compensated and decompensated eccentric hypertrophy are different answers in the wall stress and consequence rows, so the row needs both states.

Common misconceptions

· Pairing regurgitation with concentric hypertrophy. Regurgitation is a volume load and adds sarcomeres in series.

· Assuming eccentric hypertrophy always means a weak heart. While the thickness to radius ratio holds, wall stress is near normal and stroke volume is large.

Where it is taught: Section C04, the concentric against eccentric remodeling table of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C05-BD1

Brain dump 1 · foundational retrieval: Draw the baroreceptor reflex arc

A competent response contains

· Baroreceptors are stretch-sensitive nerve endings in the adventitia of the carotid sinus, a dilation at the origin of the internal carotid artery, and of the aortic arch.

· Carotid sinus afferents run in the carotid sinus nerve, the nerve of Hering, to the glossopharyngeal nerve (CN IX); aortic arch afferents run in the vagus (CN X).

· Both end in the nucleus tractus solitarius of the dorsal medulla, and the afferents release glutamate there.

· A rise in pressure raises vagal output and lowers sympathetic output; a fall does the opposite.

· The nucleus tractus solitarius excites vagal preganglionic neurons in the nucleus ambiguus and the dorsal motor nucleus of the vagus, and through the caudal ventrolateral medulla it inhibits the sympathetic premotor neurons of the rostral ventrolateral medulla that drive the intermediolateral cell column.

· Effectors: SA node (rate), AV node (conduction), ventricular myocardium (contractility, sympathetic only since the vagus has little direct effect on ventricular force), arterioles through alpha-1 receptors (resistance), veins (venous tone and MSFP), adrenal medulla (catecholamine release), and juxtaglomerular cells through beta-1 receptors (renin release).

· Carotid receptors are nearly silent below about 50 to 60 mmHg and fire maximally near 180 mmHg, with the steepest response around normal MAP, and firing also rises with the rate of pressure change.

· Aortic receptors operate about 30 mmHg higher than carotid receptors.

· The receptors reset within 1 to 2 days, so the reflex corrects short-term changes only; long-term pressure depends on renal control of volume.

Distinctions that matter

· Carotid sinus afferents travel in CN IX and aortic arch afferents in CN X, and the two sites have different operating ranges.

· Arterial baroreceptors sense pressure, while atrial and pulmonary low-pressure receptors sense volume and act mainly through the vagus on heart rate, renal sympathetic activity and ADH release.

Common misconceptions

· Treating the baroreflex as the long-term controller of blood pressure. Resetting within 1 to 2 days means it buffers change rather than setting the level.

· Expecting vagal activity to reduce ventricular contractility. Vagal innervation of ventricular muscle is sparse, so contractility is a sympathetic effect.

Where it is taught: Section C05, paragraphs on sensors and afferent pathways and on central integration and effectors of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C05-BD2

Brain dump 2 · integration and application: Run the reflex forwards and backwards in four situations

A competent response contains

· Blood loss of about 10%: firing falls, vagal output falls, sympathetic output rises, so heart rate, contractility, arteriolar tone and venous tone all rise and MAP is held nearly unchanged.

· Arteriolar constriction in that patient is concentrated in skin, gut and kidney, which gives cool pale skin and low urine output, and the fall in stroke volume narrows the pulse pressure.

· An alpha-1 vasoconstrictor drug: MAP rises, firing rises, vagal output rises and sympathetic output falls, so heart rate and contractility fall, which is reflex bradycardia, while arteriolar tone stays high because the drug is acting directly on the vessels.

· Carotid sinus massage: stretching the receptors mimics a rise in pressure, so firing rises, vagal output rises and sympathetic output falls; the SA node slows, AV nodal conduction lengthens, and arterioles dilate, so MAP falls.

· Slowed AV nodal conduction is why carotid pressure can interrupt a tachycardia whose circuit runs through the AV node.

· The older man with the tight collar has carotid sinus hypersensitivity, so pressure on the neck or head turning produces an exaggerated vagal response with bradycardia and syncope.

· Below a MAP of about 80 mmHg, peripheral chemoreceptors add sympathetic drive; below about 60 mmHg the brainstem ischemic response produces an intense sympathetic discharge.

Distinctions that matter

· A drug that raises pressure directly produces a reflex fall in heart rate, while a drug that lowers pressure directly produces a reflex rise, so the heart rate change reports the reflex and not the drug.

· Carotid massage and a fall in arterial pressure drive the same loop in opposite directions.

Common misconceptions

· Expecting arterial pressure to fall measurably after a loss of about 10% of blood volume. The reflex holds it, and tachycardia with cool skin and low urine output appears first.

· Reading reflex bradycardia after a vasoconstrictor as the drug acting on the heart. It is the baroreflex responding to the higher pressure.

Where it is taught: Section C05, paragraphs on predicting the response and on carotid sinus massage of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C05-RT01

Retrieval exercise: Rebuild the reflex response to a fall in pressure

A competent response contains

· Step 1: arterial pressure falls.

· Step 2: carotid sinus and aortic arch stretch decreases.

· Step 3: afferent firing in CN IX and CN X to the nucleus tractus solitarius falls.

· Step 4: vagal output falls and rostral ventrolateral medulla sympathetic output rises.

· Step 5: heart rate, contractility, arteriolar resistance and venous tone rise.

· Step 6: CO and TPR rise, so MAP returns toward its set point.

· Carotid sinus afferents travel in CN IX and aortic arch afferents in CN X, and both reach the nucleus tractus solitarius.

· The four effector variables that rise are heart rate, contractility, arteriolar resistance and venous tone.

· This is negative feedback because the response opposes the original change: the rising pressure restores baroreceptor firing and switches the response off.

· Reversed for a rise in pressure: stretch increases, afferent firing rises, vagal output rises and sympathetic output falls, so heart rate, contractility, arteriolar resistance and venous tone fall, CO and TPR fall, and MAP returns toward its set point.

Distinctions that matter

· The loop closes at the receptor, which is why restored pressure is what ends the response rather than any separate off switch.

· Falling pressure works through both limbs at once, withdrawing vagal tone and adding sympathetic tone.

Common misconceptions

· Drawing the pathway as sympathetic only. Withdrawal of vagal tone is the faster part of the heart rate response.

· Writing that reduced baroreceptor firing inhibits the rostral ventrolateral medulla. Reduced firing removes the inhibition, so sympathetic output rises.

Where it is taught: Section C05, the baroreceptor reflex cycle diagram of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C06-BD1

Brain dump 1 · foundational retrieval: Build the volume hormone table from memory

A competent response contains

· Angiotensin II: formed from angiotensin I by angiotensin-converting enzyme, stimulated by renin release, acting at AT1 receptors through Gq, causing arteriolar and efferent arteriolar constriction, proximal Na+/H+ exchange, aldosterone and ADH release, thirst and norepinephrine release.

· Aldosterone: from the zona glomerulosa, stimulated by angiotensin II and by a high plasma K+, acting at the intracellular mineralocorticoid receptor through gene transcription, raising ENaC and Na+/K+-ATPase so Na+ reabsorption rises and K+ and H+ secretion rise.

· ADH: from the supraoptic and paraventricular nuclei and released from the posterior pituitary, stimulated by high osmolality and by a large fall in volume or pressure, acting at V2 receptors through Gs and cAMP in the kidney and at V1 receptors through Gq in vessels, inserting aquaporin-2 for water reabsorption and constricting vessels at high levels.

· ANP: from atrial myocytes, stimulated by atrial stretch, acting at NPR-A guanylyl cyclase through cGMP, causing natriuresis, a raised GFR and vasodilation, and inhibiting renin, aldosterone and ADH.

· BNP: mainly from ventricular myocytes, stimulated by ventricular stretch and wall stress, acting through the same NPR-A and cGMP route, with the same actions, and BNP with inactive NT-proBNP marks ventricular strain.

· RAAS sequence: low perfusion pressure, low macula densa NaCl or beta-1 input; juxtaglomerular cells release renin; renin cleaves angiotensinogen to angiotensin I; pulmonary ACE forms angiotensin II; arteriolar constriction, proximal Na+ reabsorption, thirst and ADH follow; aldosterone raises ENaC and Na+/K+-ATPase in principal cells; Na+ and water retention raise volume and pressure.

· The three stimuli for renin are low renal perfusion pressure with less arteriolar stretch, low macula densa NaCl delivery, and beta-1 stimulation; angiotensin II and ANP inhibit release.

· Renin comes from juxtaglomerular cells of the afferent arteriole, and the cleavage of angiotensinogen is the rate-limiting step.

· ACE is most abundant on pulmonary capillary endothelium and also inactivates the vasodilator bradykinin.

· Angiotensin II constricts the efferent arteriole more than the afferent, preserving filtration pressure when renal perfusion is low.

· Timing: the baroreceptor reflex acts in seconds, angiotensin II in minutes, and aldosterone over hours to days because it works through new protein synthesis.

Distinctions that matter

· Aldosterone retains Na+ with water and so changes volume, while ADH retains water alone and so changes osmolality.

· ANP and BNP differ in their source and stimulus, atrial stretch against ventricular stretch and wall stress, not in their actions.

Common misconceptions

· Treating renin as the hormone that acts on vessels and nephron. Renin is an enzyme, and angiotensin II and aldosterone carry the actions.

· Expecting aldosterone to act within minutes. Its effect takes hours because it works through transcription of ENaC and Na+/K+-ATPase.

Where it is taught: Section C06, the hormonal regulators table, the RAAS paragraphs and flow diagram, and the final paragraph on timing of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C06-BD2

Brain dump 2 · integration and application: Three patients, three hormone patterns

A competent response contains

· Patient A: renin rises from low renal perfusion pressure, low macula densa NaCl delivery and beta-1 stimulation, so angiotensin II and aldosterone rise; ADH rises because the fall in volume is large enough, on the order of 10% or more, and angiotensin II also stimulates its release; ANP and BNP fall because atrial and ventricular stretch fall.

· Patient B: osmoreceptors in the anterior hypothalamus detect the rise in osmolality at a change of about 1%, so ADH rises; renin, angiotensin II and aldosterone change little because volume and renal perfusion are normal, and the natriuretic peptides change little.

· Patient C: aldosterone is high from the adenoma, so renin and angiotensin II are suppressed by the resulting volume expansion and higher pressure; ANP rises with atrial stretch; ADH is not the driver.

· Patient A retains Na+ with water through aldosterone and angiotensin II, so extracellular and plasma volume rise; patient B retains water alone through ADH and aquaporin-2 insertion in the collecting duct, which corrects osmolality without adding much volume.

· The difference follows from the target: aldosterone raises ENaC and Na+/K+-ATPase so Na+ is reabsorbed, while ADH inserts aquaporin-2 so water follows the medullary gradient.

· In patient C, Na+ retention raises pressure, and the lumen-negative potential created by Na+ reabsorption drives K+ secretion, so plasma K+ falls.

· There is no edema because within days the higher pressure and ANP increase Na+ and water excretion enough to halt the expansion, which is aldosterone escape.

· Long-term pressure control rests on renal pressure natriuresis: higher pressure increases Na+ and water excretion until volume and pressure return to the level the kidney defends.

· In heart failure natriuretic peptide levels are high but the renal response is blunted, so they cannot overcome RAAS and sympathetic drive.

· Inappropriate ADH secretion retains water rather than Na+, which dilutes plasma Na+ and causes hyponatremia rather than hypertension.

Distinctions that matter

· Osmolality needs about a 1% change to release ADH, while volume or pressure needs a change on the order of 10% or more, which is why osmoregulation dominates day to day.

· High aldosterone gives hypertension with hypokalemia and no edema, while heart failure gives high aldosterone with edema, because the kidney in heart failure is also seeing a low perfusion signal.

Common misconceptions

· Expecting high aldosterone to cause edema. Aldosterone escape halts the expansion, so pressure and K+ change but interstitial volume does not keep rising.

· Assuming high natriuretic peptide levels mean the kidney is excreting salt briskly. In heart failure the renal response to them is blunted.

Where it is taught: Section C06, the ADH and natriuretic peptide paragraphs, the final paragraph on pressure natriuresis, and the chapter integration section of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C06-PR01

Prediction exercise: Predict the response to renal artery narrowing

A competent response contains

· Renin release from the affected kidney rises; the other two stimuli at that cell are reduced stretch of the afferent arteriole from low perfusion pressure and reduced NaCl delivery to the macula densa.

· Circulating angiotensin II rises, because renin cleaving angiotensinogen to angiotensin I is the rate-limiting step and pulmonary ACE then forms angiotensin II.

· Efferent arteriolar tone rises more than afferent tone, which preserves glomerular filtration pressure while renal perfusion is low.

· Aldosterone secretion rises under angiotensin II, and transcription raises the number of apical epithelial Na+ channels and of basolateral Na+/K+-ATPase in principal cells.

· Urinary K+ and H+ secretion rise, because Na+ reabsorption leaves the lumen electrically negative, which drives K+ secretion from principal cells and H+ secretion from intercalated cells.

· Plasma volume rises as Na+ and water are retained, and systemic arterial pressure rises over the next days.

· Renin release falls in the normal kidney once systemic pressure has risen, but the affected kidney still sees a low pressure beyond the narrowing, so it keeps releasing renin and the cycle continues.

Distinctions that matter

· Low perfusion pressure at the juxtaglomerular cell is not the same as low systemic pressure, which is why renin stays high while the arm cuff reads hypertensive.

· Angiotensin II acts within minutes while aldosterone takes hours, because one binds a Gq-coupled receptor and the other works through transcription.

Common misconceptions

· Expecting the rise in systemic pressure to switch off renin from the narrowed side. The narrowing keeps the pressure at that juxtaglomerular cell low.

· Expecting glomerular filtration to collapse in the affected kidney. Preferential efferent constriction by angiotensin II holds filtration pressure up.

Where it is taught: Section C06, the renin-angiotensin-aldosterone paragraphs and flow diagram of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C07-BD1

Brain dump 1 · foundational retrieval: What standing does and how the body answers it

A competent response contains

· Orthostatic hypotension is a sustained fall in systolic pressure of at least 20 mmHg, or diastolic pressure of at least 10 mmHg, within 3 minutes of standing or of head-up tilt to at least 60 degrees.

· Standing raises foot venous pressure to about 90 mmHg in a motionless adult.

· Roughly half a litre of blood pools in the leg and pelvic veins within seconds, and plasma filters into the leg interstitium over minutes.

· Venous return falls, so SV, CO and MAP fall in turn.

· The baroreceptor reflex raises heart rate by about 10 to 20 bpm and constricts splanchnic and muscle arterioles and veins, restoring MAP within about a minute with systolic pressure slightly lower and diastolic pressure slightly higher.

· The skeletal muscle pump drives blood upward, and venous valves make that flow one way.

· Renin and ADH add support over minutes.

· Symptoms appear once MAP at head level falls below the lower limit of cerebral autoregulation, about 60 mmHg: lightheadedness, blurred vision, weakness, cognitive slowing and syncope.

· A coat-hanger ache of the neck and shoulders is the pattern specific to the neurogenic forms.

Distinctions that matter

· Venous pooling happens in seconds while plasma filtration into the leg tissue takes minutes, so the two phases of the fall have different time courses.

· The pressure criteria define the sign, and the symptoms depend on cerebral perfusion, so a patient can meet the criteria without symptoms if autoregulation holds.

Common misconceptions

· Reading a transient dip on standing as orthostatic hypotension. The definition requires a sustained fall within 3 minutes.

· Expecting both systolic and diastolic criteria to be met. Either one is enough.

Where it is taught: Section C07, the definition paragraph and the paragraph on why standing threatens arterial pressure of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C07-BD2

Brain dump 2 · integration and application: Separate the mechanisms from two sets of vital signs

A competent response contains

· Patient A: systolic fall is 112 minus 88, which is 24 mmHg, and the heart rate rise is 128 minus 92, which is 36 bpm; the definition is met, the reflex arc is intact, and the mechanism is volume depletion.

· Patient B: systolic fall is 162 minus 118, which is 44 mmHg, and the heart rate rise is 100 minus 96, which is 4 bpm; the definition is met, the afferent and central limbs are working but the efferent sympathetic limb is not, and the mechanism is neurogenic orthostatic hypotension from autonomic failure.

· The heart rate rise is the single number that separates them: a large rise means an intact reflex working against too little volume, and little or no rise means the efferent sympathetic pathway cannot release enough norepinephrine.

· Patient B is hypertensive lying down partly because denervated vessels are hypersensitive to circulating pressors, and supine hypertension is common in autonomic failure.

· In long-standing diabetes the nerve injury mechanisms are aldose reductase converting glucose to sorbitol with depletion of NADPH, which leaves less glutathione regeneration and more oxidative stress, accumulation of advanced glycation end products, and vasa nervorum disease adding ischemia.

· Long fibres fail first and vagal loss often precedes sympathetic loss, so a fixed resting tachycardia appears before splanchnic vasoconstrictor loss causes orthostatic hypotension.

· Postural tachycardia syndrome is a heart rate rise of at least 30 bpm in adults on standing with symptoms but no significant fall in pressure.

· In vasovagal syncope sympathetic tone is suddenly withdrawn while vagal output surges, so pressure and heart rate fall together.

· Volume depletion also arises from hemorrhage, vomiting, diarrhoea, diuresis and burns, and from adrenal insufficiency, where aldosterone loss causes renal Na+ wasting and cortisol loss reduces vascular responsiveness to catecholamines.

Distinctions that matter

· A fall in pressure with a brisk heart rate rise is a volume problem, while a larger fall with a flat heart rate is an efferent nerve problem.

· Postural tachycardia syndrome has the tachycardia without the pressure fall, and vasovagal syncope has both falling together, so neither matches either of these patients.

Common misconceptions

· Taking the size of the pressure fall as the discriminator. The heart rate response is what separates the mechanisms.

· Assuming autonomic failure means low pressure at all times. Supine hypertension is common in these patients.

Where it is taught: Section C07, the mechanisms paragraphs, the table distinguishing the main mechanisms, and the worked example on two sets of orthostatic vital signs of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C07-CA01

Clinical application: Lightheadedness on standing after two weeks in bed

A competent response contains

· Recognize: systolic pressure fell 128 minus 96, which is 32 mmHg, diastolic fell 76 minus 60, which is 16 mmHg, and the heart rate rose 104 minus 78, which is 26 bpm at 3 minutes standing, so both pressure criteria are met within the time limit and this is orthostatic hypotension.

· Explain: prolonged bed rest shifts blood toward the thorax and stretches the atria, so ANP rises and ADH falls, and the resulting diuresis lowers plasma volume within days; the ventricle becomes smaller and stiffer, baroreflex sensitivity falls, and the leg muscle pump weakens with disuse.

· Differentiate: volume loss through vomiting or bleeding gives the same intact reflex pattern but with an external route of loss and usually a raised urea to creatinine ratio and thirst, whereas this patient lost volume by renal excretion during recumbency; failure of the sympathetic efferent pathway would show little or no heart rate rise with supine hypertension, and the 26 bpm rise here is what separates this patient from that.

· Investigate: a heart rate rise on standing of roughly 20 bpm or more, confirming an intact efferent limb; a supine pressure that is normal rather than high, here 128/76 mm Hg; a urea to creatinine ratio within the reference range, here 14, arguing against brisk external volume loss; a normal plasma Na+ at 139 mEq/L, since water was lost with Na+ rather than water alone; the weight loss of 3 kg and the history of 15 days of immobility; and the coat-hanger ache, which raises neurogenic forms and is checked against the heart rate response.

· Decide: three of the four defences are weakened, since plasma volume and MSFP are low, baroreflex sensitivity is reduced, and the muscle pump is deconditioned, while the reflex arc itself, the afferent, central and efferent sympathetic pathway, remains intact.

· Justify: the lower plasma volume reduces stressed volume, so MSFP falls and the venous return curve shifts left, which moves the operating point to a lower CO at a lower RAP; pooling of roughly half a litre on standing shifts it further left, and because venous return sets what the heart can eject, the intact reflex can raise heart rate, contractility and vascular tone but cannot create the filling volume that the stiffer, smaller ventricle needs, so MAP falls despite a brisk heart rate rise.

Distinctions that matter

· An intact reflex with a brisk heart rate rise separates a filling problem from autonomic failure, and the supine pressure separates them again.

· Deconditioning weakens both the volume and the pump, while the nerve pathway stays intact, which is why the pattern looks like volume depletion without any external fluid loss.

Common misconceptions

· Reading the normal urea to creatinine ratio and normal plasma Na+ as evidence against volume depletion. Recumbency-induced diuresis removes Na+ with water, so osmolality and the ratio can stay normal.

· Attributing the coat-hanger ache alone to autonomic failure here. The heart rate rise of 26 bpm shows the efferent sympathetic limb is working.

Where it is taught: Section C07, mechanisms paragraph on prolonged bed rest and the mechanism table, with the venous return reasoning in C03 of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C01-Q01

Board-style question: Reading the arteriovenous oxygen difference

Correct answer: C. It has fallen from about 5.0 L/min to about 3.1 L/min, because each litre of blood now gives up more oxygen

Why it is right: With VO2 fixed at 250 mL/min, output varies inversely with the arteriovenous O2 difference, so widening it from 50 to 80 mL per litre lowers calculated output from 5.0 to about 3.1 L/min, and the low mixed venous content shows tissues extracting more from each litre because delivery has fallen.

A is wrong: A rise in output would narrow the arteriovenous difference, since less extraction per litre is needed, and extraction here has clearly increased rather than decreased.

B is wrong: Output cannot be unchanged when the denominator of the Fick relationship has changed while the numerator and arterial content have not; the mixed venous value is what carries the information.

D is wrong: Mixed venous content is the second term of the content difference in the Fick relationship, so sampling it from the pulmonary artery is exactly what makes the estimate possible.

A competent response contains

· Fick: CO = VO2 / (CaO2 - CvO2), with content differences converted from mL/dL to mL per litre by multiplying by 10.

· Last year the difference was 20 minus 15, which is 5 mL/dL or 50 mL/L, so CO was 250 divided by 50, which is 5.0 L/min.

· Today the difference is 20 minus 12, which is 8 mL/dL or 80 mL/L, so CO is 250 divided by 80, which is about 3.1 L/min.

· A widening arteriovenous O2 difference at an unchanged VO2 means output has fallen and tissues are extracting more O2 from each litre of blood.

Distinctions that matter

· A falling mixed venous content reports greater extraction, which follows from reduced delivery rather than from any change in arterial oxygenation.

Common misconceptions

· Reading a wider arteriovenous difference as better oxygen delivery. It means each litre is giving up more because fewer litres are arriving.

Where it is taught: Section C01 of the CV-6 Cardiac Output and Its Regulation notes.

NPLEX-P1-CARD-06-01-C04-Q02

Board-style question: Wall stress in a dilating ventricle

Correct answer: A. Systolic wall stress has risen in proportion to the radius, because wall stress varies with pressure times radius divided by twice the thickness

Why it is right: In the law of Laplace wall stress rises in direct proportion to chamber radius at a fixed pressure and thickness, so a 40% larger radius raises systolic wall stress by about 40%, and systolic wall stress is one of the three main determinants of myocardial O2 demand.

B is wrong: Myocardial O2 extraction at rest is already about 70 to 75%, not about 25%, and it is extraction being near maximal that forces demand to be met by coronary flow instead.

C is wrong: A larger radius raises systolic wall stress rather than lowering it, and pressure work costs far more O2 than volume work, so the reasoning is reversed on both counts.

D is wrong: Most left ventricular coronary flow occurs in diastole rather than systole, and coronary flow is the supply side of the balance, not the source of the O2 cost of a beat.

A competent response contains

· Wall stress = (P x r) / (2 x h), so with pressure and thickness unchanged a 40% larger radius raises systolic wall stress by about 40%.

· Systolic wall stress is one of the three main determinants of myocardial O2 demand, alongside heart rate and contractility.

· The dilated ventricle must develop more force per unit of wall to produce the same pressure, so each beat costs more O2.

Distinctions that matter

· Radius and thickness act on wall stress in opposite directions, so dilation without matching thickening raises the O2 cost of every beat.

Common misconceptions

· Assuming a larger chamber shares the load better. A larger radius raises wall stress in direct proportion.

Where it is taught: Section C04 of the CV-6 Cardiac Output and Its Regulation notes.